Mars Habitat

Campaign 1 - Case 03 - Arcadia Planitia, Mars

SOLAR
AGRICULTURAL
AUTHORITY

Name ______________________ **Date** ______________ **Period** ______

Mission question

How can the habitat receive an adequate total amount of light while potato plants still fail to form normal green new growth?

CREW
Top-canopy and newest leaves bleach after three weeks; added iron and nitrogen did not help.

SENSORS
Combined PPFD is 280 umol m-2 s-1, described as adequate; primary path is a 12 m light pipe plus white LEDs.

PLANTS
Older lower leaves retain green; new tissue is pale yellow-white; roots are healthy.

LOGS
Spectral analysis was skipped because total PAR was considered sufficient.

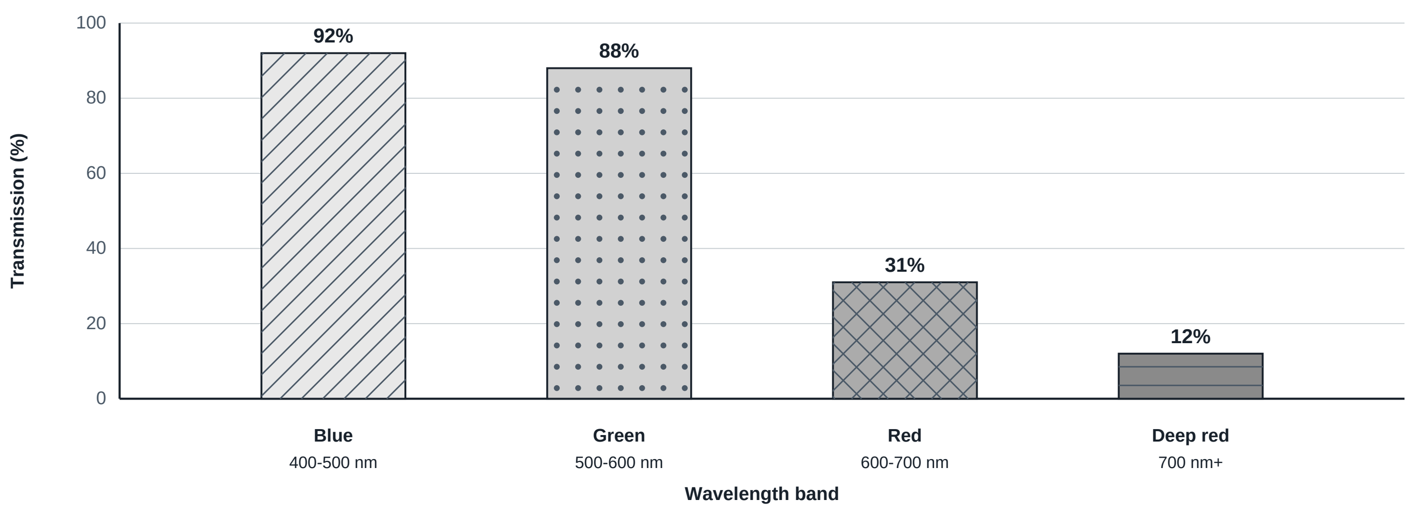
TASK 1 - Define the measurement

State what the 280 PPFD reading measures and one important feature it cannot show.

TASK 2 - Read the spectral-transmission data

Read the four exact runtime values. Identify the lowest transmission inside 400-700 nm. Do not invent intermediate measurements.

Surface intake to habitat output transmission



SOURCE STATUS: Whole-number transmission values are game-provided. The graph does not imply a continuous measured spectrum.

LOWEST IN-PAR TRANSMISSION AND EVIDENCE

TASK 3 - Compare quantity and quality

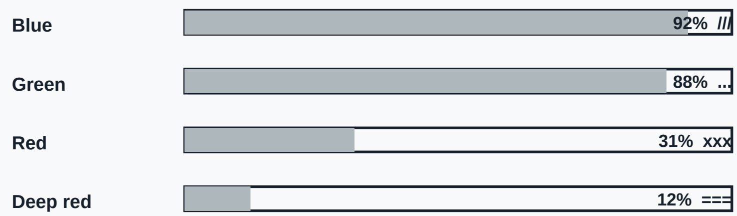
Compare the adequate total PPFD reading with the wavelength distribution. Explain why “not enough total light” does not fit the measured quantity.

Quantity check

280 umol m-2 s-1 combined PPFD

Runtime status: adequate total photon quantity

Spectral transmission by band



Analytical conclusion: adequate total does not prove an adequate wavelength distribution.

TASK 4 - Connect the symptom pattern

Use the old-versus-new leaf pattern, healthy roots, and failed nutrient additions. Explain why the evidence points to a problem in forming new chlorophyll.

Observation	What it supports
Older leaves retain green	Existing pigment was already present.
Newest tissue becomes pale first	New pigment formation is the affected step.
Roots are healthy	Root poisoning is not supported.
Iron and nitrogen did not help	Simple nutrient shortage is weakened.

TASK 5 - Select and reject diagnoses

Mark the diagnosis that fits all evidence. Then reject one tempting alternative with a specific observation or measurement.

Select	Diagnosis
[ ]	Mars soil perchlorates are poisoning the roots.
[ ]	CO_2 concentration is too high.
[ ]	The Mars sol is disrupting the photoperiod.
[ ]	The light-delivery system filters red wavelengths needed for chlorophyll biosynthesis.

REJECTED ALTERNATIVE AND EVIDENCE

TASK 6 - Model the mechanism

Complete the middle steps. Your chain must connect wavelength-selective transmission to the location of the symptom.

47-sol filter replacement → **Complete filter problem** → **Complete spectral result** → **Complete pigment result** → **Pale new growth**

Case mechanism model. General plant-light responses also involve blue, green, and red wavelengths.

MECHANISM EXPLANATION

SCIENCE BOUNDARY: This case isolates a red-band rejection. Plants do not use only red light; blue, green, and red wavelengths can all affect plant function.

TASK 7 - Write the case conclusion

Write a concise Claim-Evidence-Reasoning conclusion. Use the 280 PPFD value, at least two spectral values, and the symptom pattern.

Claim

Evidence

Reasoning

TASK 8 - Transfer the analysis

Explain why making the system brighter without correcting the spectrum might fail. Name the next measurement or engineering check.

TASK 9 - Exit ticket

In a new controlled-environment lighting failure, what two measurements would you compare first, and why?